

Optimal Control of Photovoltaic Battery Storage

Ajith George¹ Ashna Benny¹ Umair Ahmad¹

¹Friedrich-Alexander Universität Erlangen-Nürnberg

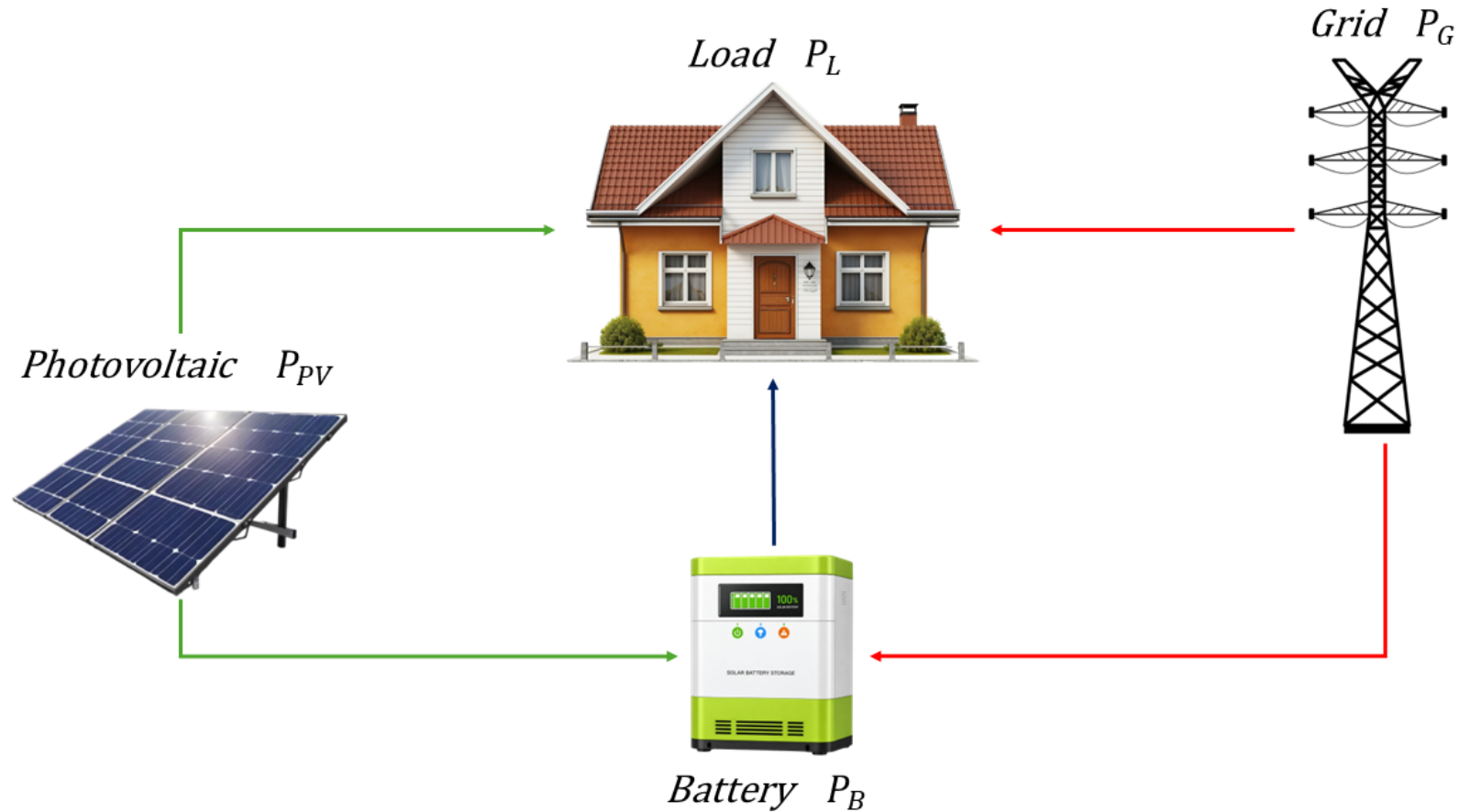
June 12, 2026

Motivation

- Solar energy is expanding worldwide
- Solar generation is intermittent
- Mismatch in demand and generation
- Need for efficient energy management
- Battery storage enables energy balancing
- Goal: Reduce electricity cost

System Description

System Components & Energy Flow Concept



Time Discretization

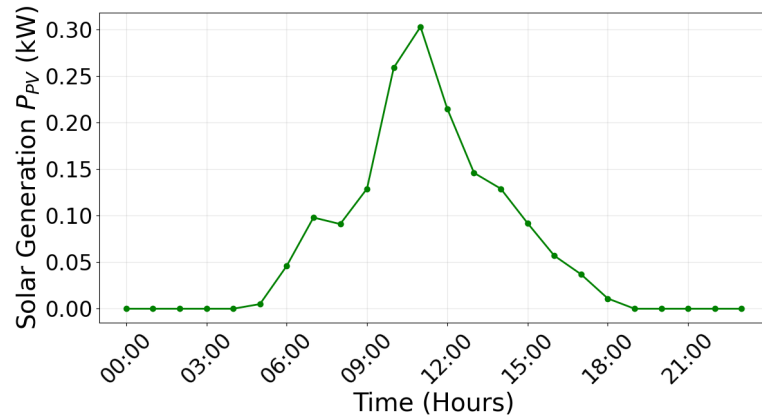
- Time horizon: *24 hours*
- Time interval : *60 minutes*
- Total steps : $i = 1, 2, \dots, 24$
- Converts continuous problem into a **finite-dimensional optimization problem**

Initial Modeling Approach

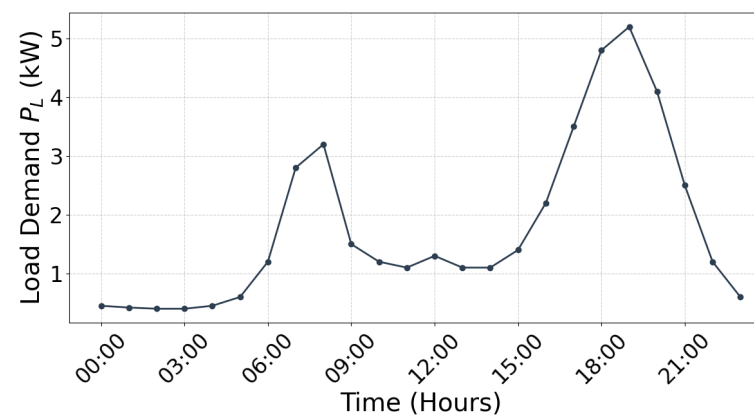
Model Inputs

- $P_{PV}(i)$: Solar generation
- $P_L(i)$: Load demand
- $P_C(i)$: Electricity price

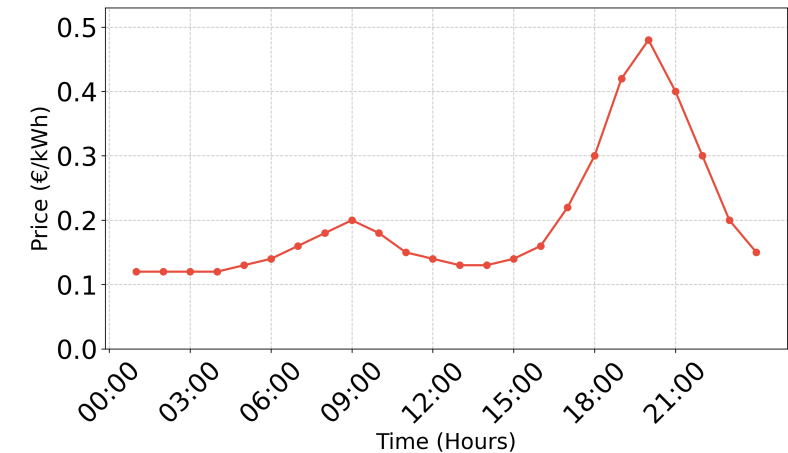
(All assumed **known from forecasts**)



Solar generation $P_{PV}(i)$



Load demand $P_L(i)$



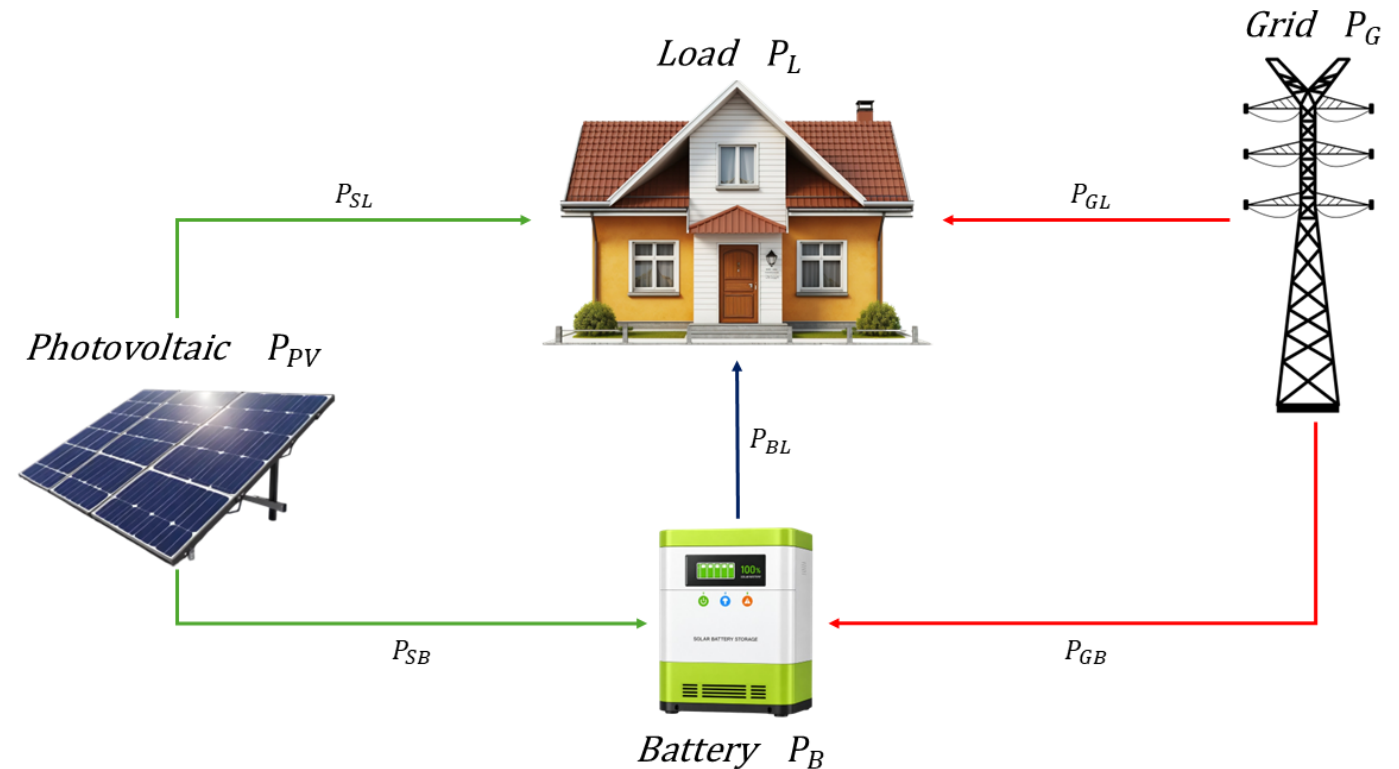
Electricity price $P_C(i)$

Data sources: PV generation from Renewables.ninja; electricity price data from ENTSO-E Transparency Platform; load profile assumed (synthetic). Data corresponds to May 8, 2019.

Initial Modeling Approach

Decision Variables

- Grid Supplies:
 - $P_{GB}(i) : \text{Grid} \rightarrow \text{Battery}$
 - $P_{GL}(i) : \text{Grid} \rightarrow \text{Load}$
- Battery supplies:
 - $P_{BL}(i) : \text{Battery} \rightarrow \text{Load}$
- Photovoltaic Supplies:
 - $P_{SL}(i) : \text{Solar} \rightarrow \text{Load}$
 - $P_{SB}(i) : \text{Solar} \rightarrow \text{Battery}$



Constraints

- Power Balance (Load Satisfaction):

$$P_{BL}(i) + P_{GL}(i) + P_{SL}(i) = P_L(i)$$

- Solar Generation Constraint:

$$P_{SB}(i) + P_{SL}(i) \leq P_{PV}(i)$$

- Battery Capacity:

$$B_{min} \leq B(i) \leq B_{max}$$

- Battery Dynamics:

$$B(i+1) = B(i) + P_{GB}(i) + P_{SB}(i) - P_{BL}(i)$$

$$\forall i = 1, 2, \dots, 24$$

Full Model

- Minimize total electricity cost
- Reduce expensive grid usage

Optimization Problem:

$$\min_{P_{BL}, P_{SL}, P_{GB}, P_{SB}, [P_{GL}]} \sum_{i=1}^{24} P_C(i) \cdot (P_{GL}(i) + P_{GB}(i))$$

such that

$$P_{BL}(i) + P_{GL}(i) + P_{SL}(i) = P_L(i)$$

$$P_{SB}(i) + P_{SL}(i) \leq P_{PV}(i)$$

$$B_{min} \leq B(i) \leq B_{max}$$

$$B(i+1) = B(i) + P_{GB}(i) + P_{SB}(i) - P_{BL}(i)$$

$$\forall i = 1, 2, \dots, 24$$

Given:

$$P_L, P_{PV}, P_C \in \mathbb{R}^{24} \text{ and } B_{min}, B_{max} \in \mathbb{R}$$

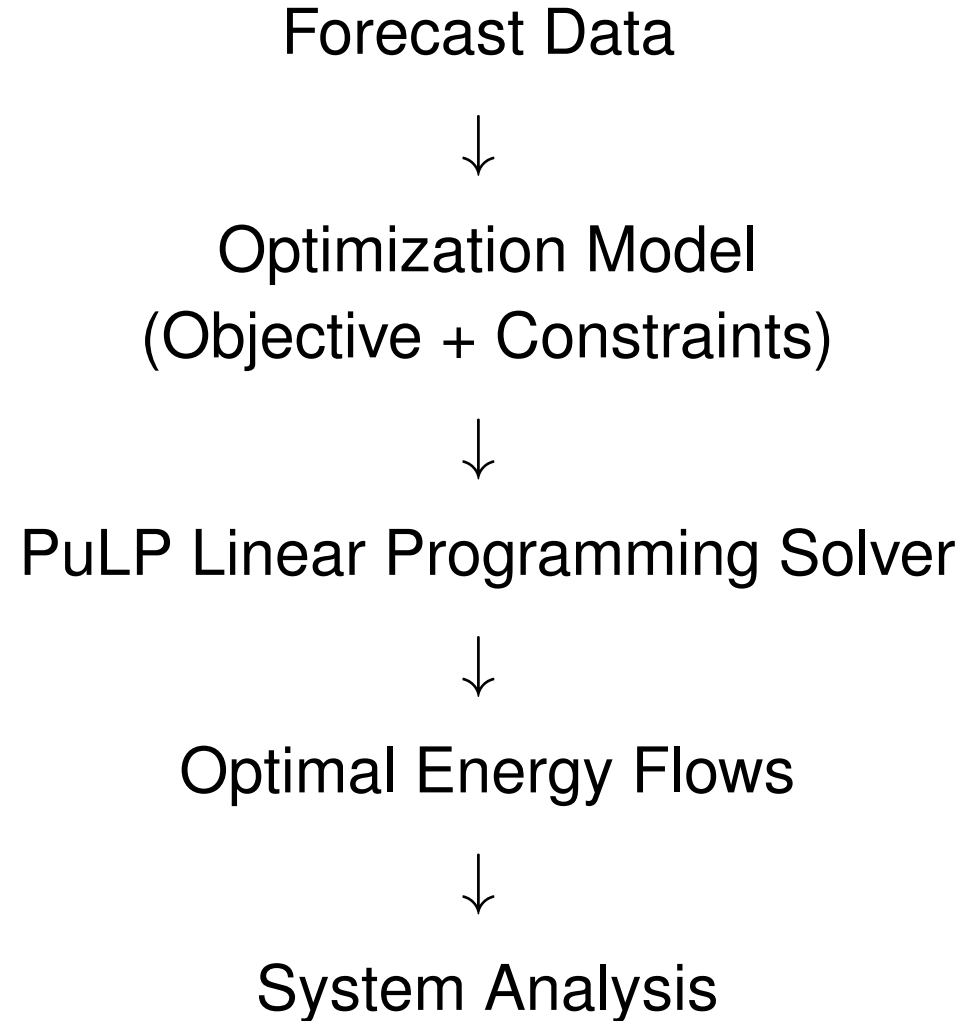
Model Type

- Deterministic optimization model
- Linear formulation
- Inputs assumed known (forecast-based)
- Finite-dimensional problem

- Phase 1: Modeling & Formulation
 - System definition and model setup
 - Deterministic linear model
- Phase 2: Solution
 - Implement in Python
 - Solve optimization problem
- Phase 3: Analysis
 - Battery behavior, grid vs solar usage
 - Cost evaluation
- Phase 4: Extension
 - Stochastic optimization
 - Robust optimization
- Phase 5: Evaluation
 - Compare results
 - Discuss improvements

1. Phase 2: Solution

2. Phase 3: Analysis

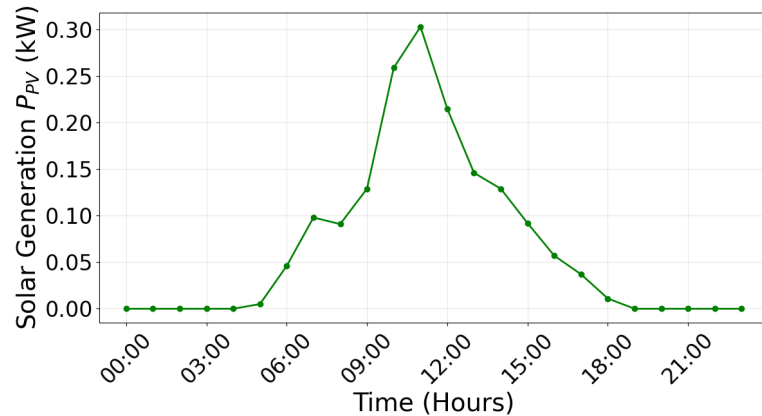


Defining the Daily Scenario

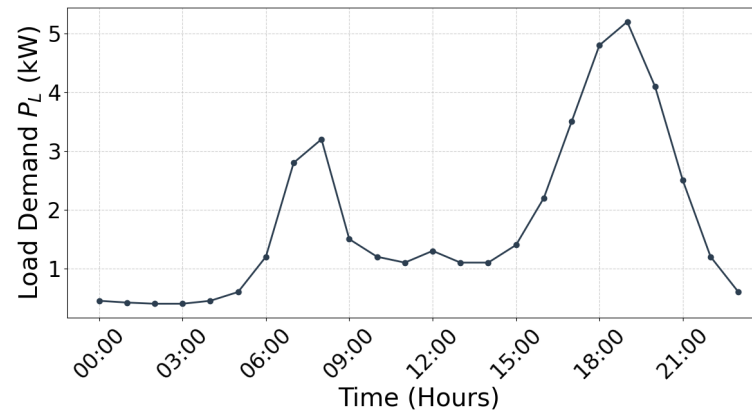
Phase 2: Solution

- **Load Profile (P_L):** Variable household demand, featuring a significant evening peak of 5.20 kW at hour 20.
- **Solar Profile (P_{PV}):** Standard daytime generation curve peaking during midday hours.
- **Dynamic Pricing (C):** Highly variable grid tariffs to encourage arbitrage.
 - *Off-peak (Night):* €0.12 / kWh
 - *Peak (Evening):* €0.48 / kWh

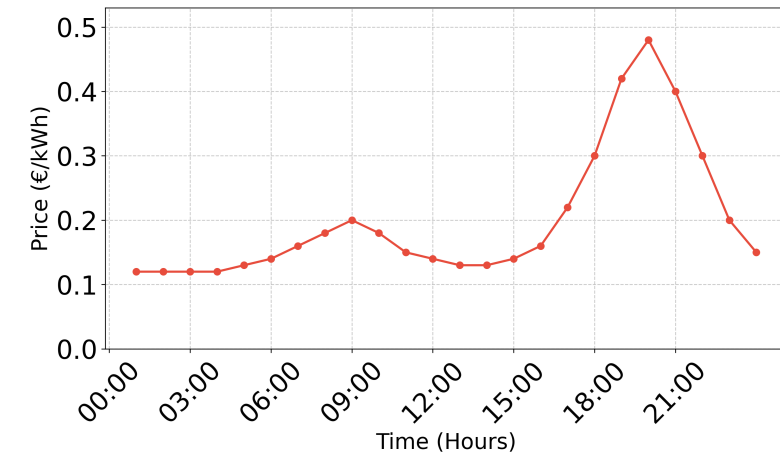
(All assumed **known from forecasts**)



Solar generation $P_{PV}(i)$



Load demand $P_L(i)$



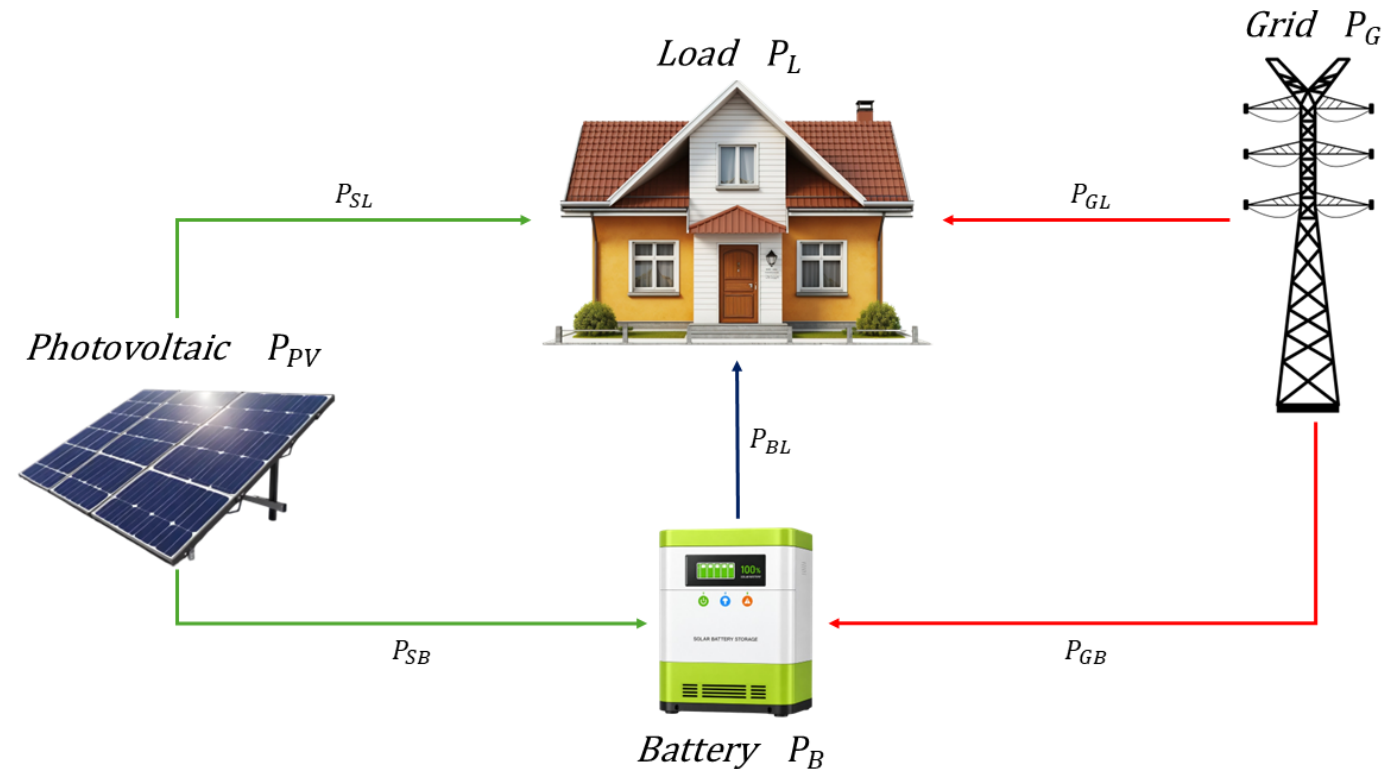
Electricity price $P_C(i)$

Data sources: PV generation from Renewables.ninja; electricity price data from ENTSO-E Transparency Platform; load profile assumed (synthetic). Data corresponds to May 8, 2019.

Initial Modeling Approach

Decision Variables

- Grid Supplies:
 - $P_{GB}(i) : \text{Grid} \rightarrow \text{Battery}$
 - $P_{GL}(i) : \text{Grid} \rightarrow \text{Load}$
- Battery supplies:
 - $P_{BL}(i) : \text{Battery} \rightarrow \text{Load}$
- Photovoltaic Supplies:
 - $P_{SL}(i) : \text{Solar} \rightarrow \text{Load}$
 - $P_{SB}(i) : \text{Solar} \rightarrow \text{Battery}$



Full Model

- Minimize total electricity cost
- Reduce expensive grid usage

Optimization Problem:

$$\min_{P_{BL}, P_{SL}, P_{GB}, P_{SB}, [P_{GL}]} \sum_{i=1}^{24} P_C(i) \cdot (P_{GL}(i) + P_{GB}(i))$$

such that

$$P_{BL}(i) + P_{GL}(i) + P_{SL}(i) = P_L(i)$$

$$P_{SB}(i) + P_{SL}(i) \leq P_{PV}(i)$$

$$B_{min} \leq B(i) \leq B_{max}$$

$$B(i+1) = B(i) + [P_{GB}(i) + P_{SB}(i)] * 0.90 - [P_{BL}(i)]/0.95$$

$$\forall i = 1, 2, \dots, 24$$

Given:

$$P_L, P_{PV}, P_C \in \mathbb{R}^{24} \text{ and } B_{min}, B_{max} \in \mathbb{R}$$

Phase 2: Solution

- **Framework:** Built the Linear Programming (LP) model using the PuLP library in Python.
- **Mathematical Solver:** Utilized the default CBC (COIN-OR Branch and Cut, Simplex).
- **Time Discretization:** Modeled a full 24-hour horizon with 1-hour intervals ($\Delta t = 1$ hour).
- **Objective:** Minimize total daily grid electricity costs while strictly satisfying household load and hardware limits.

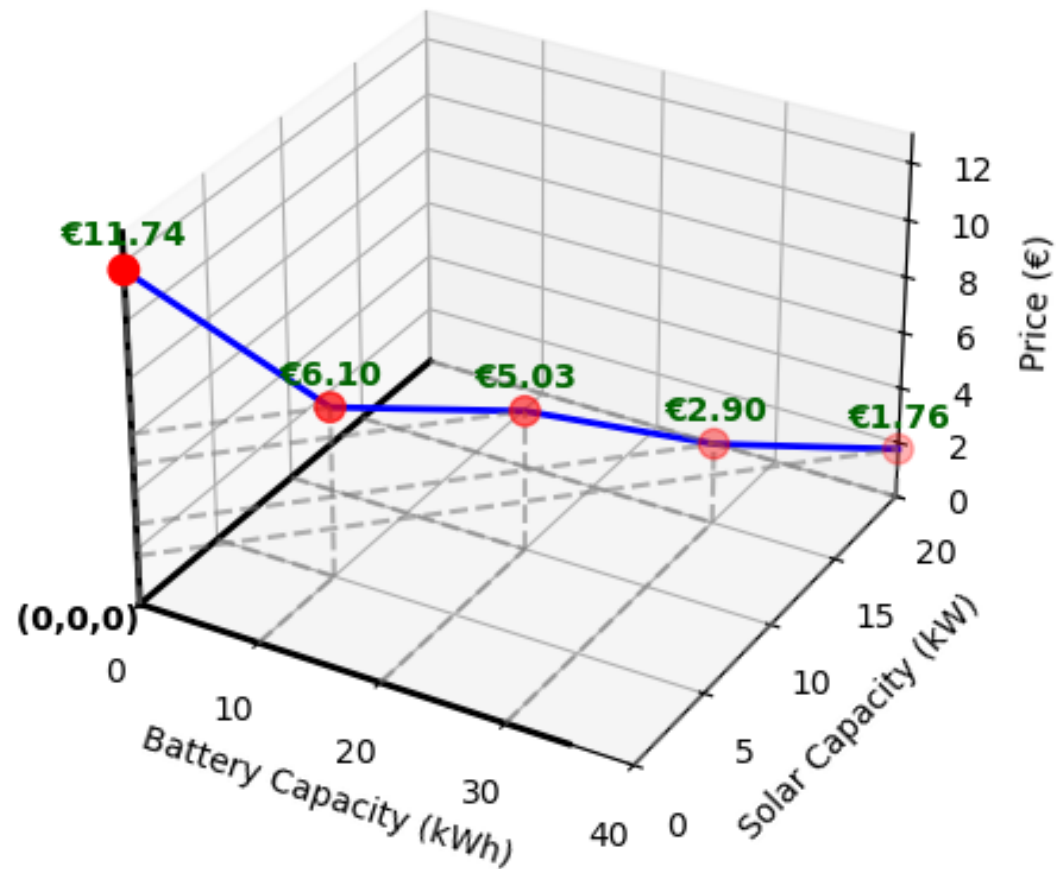
Phase 2: Solution

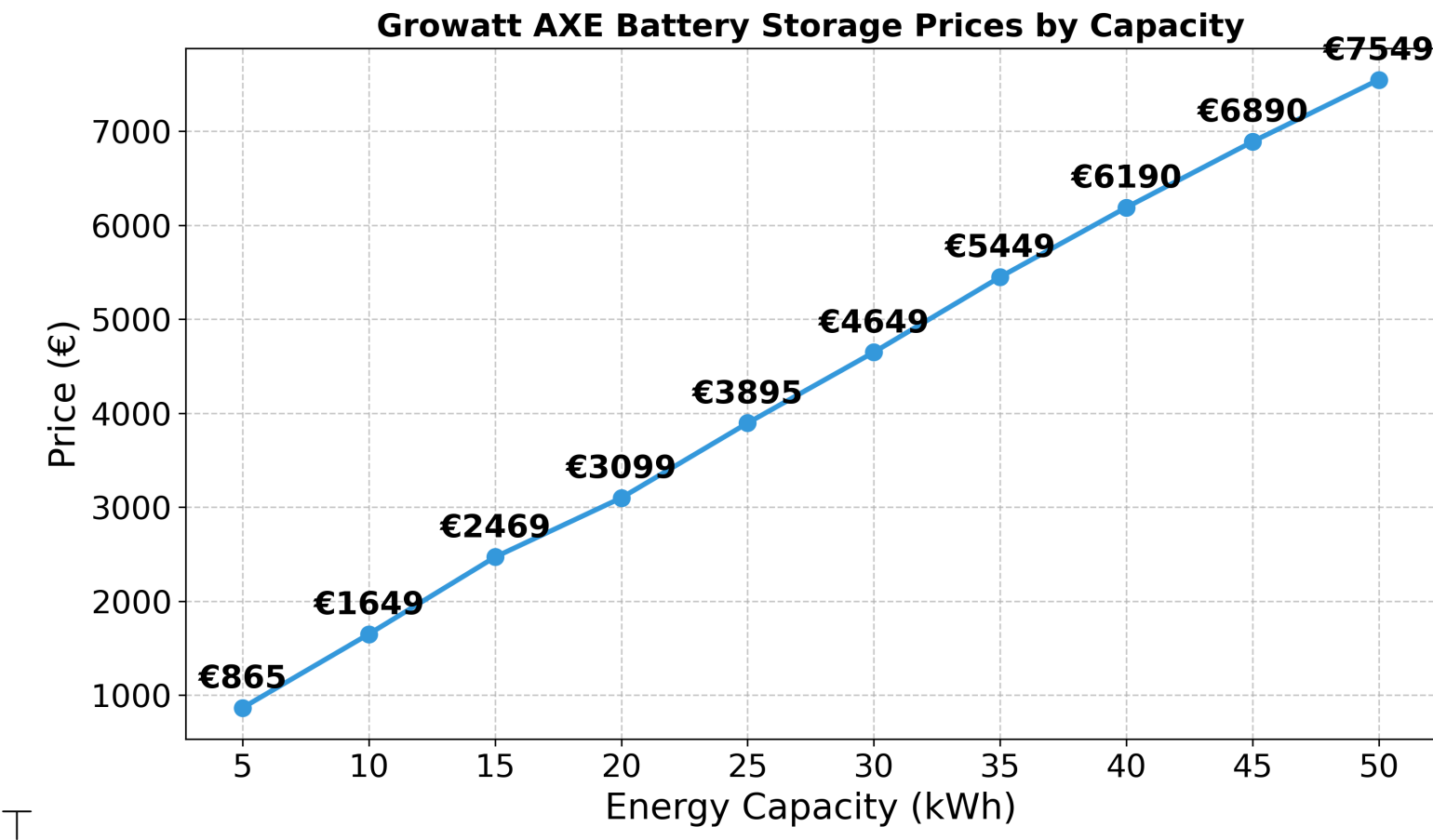
- **Capacity:** Battery capacity of 30.0 kWh.
- **Safety Limits:** Restricted State of Charge (SoC) to protect battery health:
 - $B_{min} = 20\%$ (6.0 kWh)
 - $B_{max} = 90\%$ (27.0 kWh)
- **Efficiency:** Applied realistic efficiency losses ($\eta_{charge} = 0.90$, $\eta_{discharge} = 0.95$).

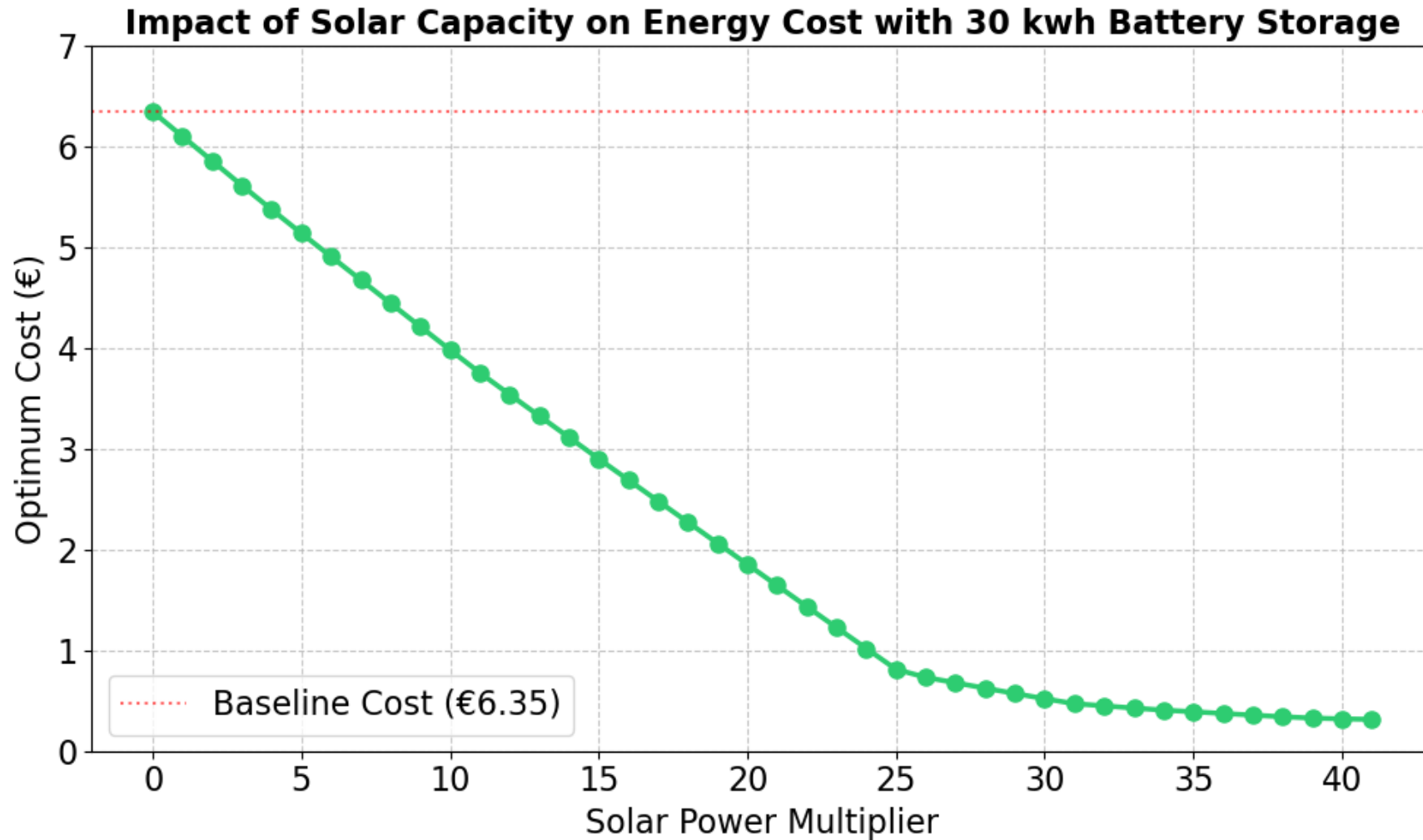
Parameter	Value
Time Horizon	24 h
Time Step	1 h
Solar power	25 kw
Battery Capacity	30 kWh
Minimum SoC	20%
Maximum SoC	90%
Charging Efficiency	90%
Discharging Efficiency	95%
Solver	PuLP (LP)

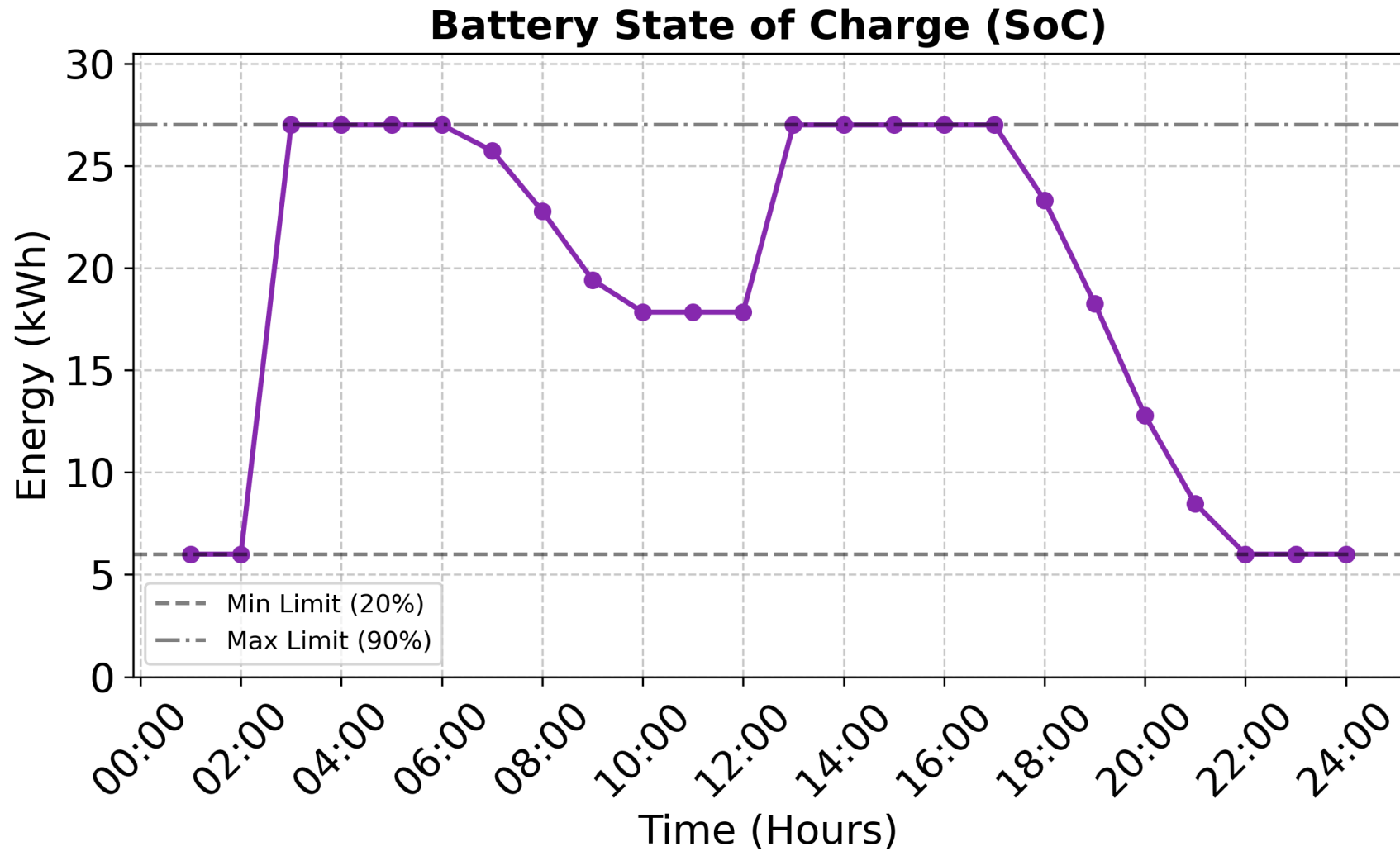
1. Phase 2: Solution

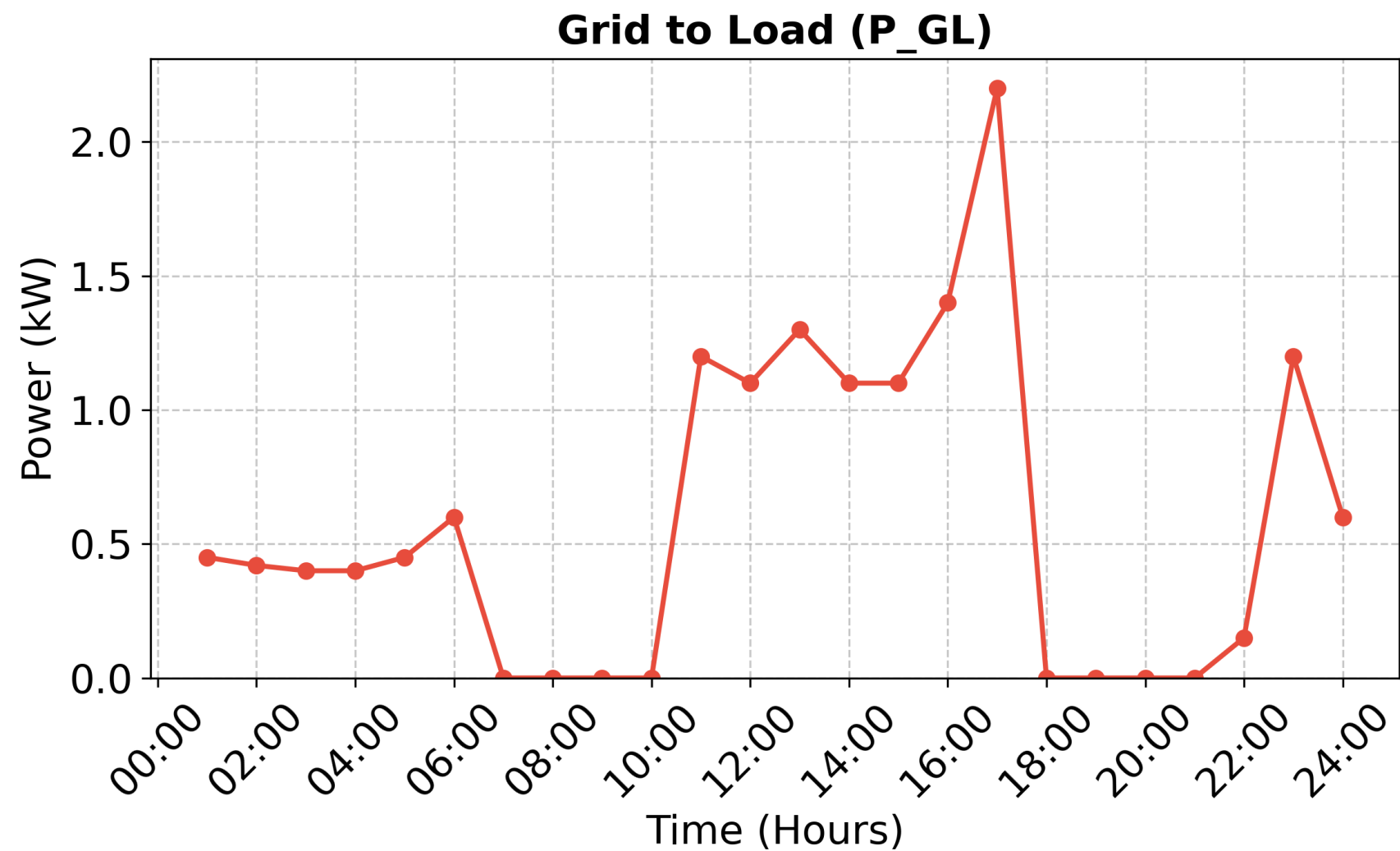
2. Phase 3: Analysis

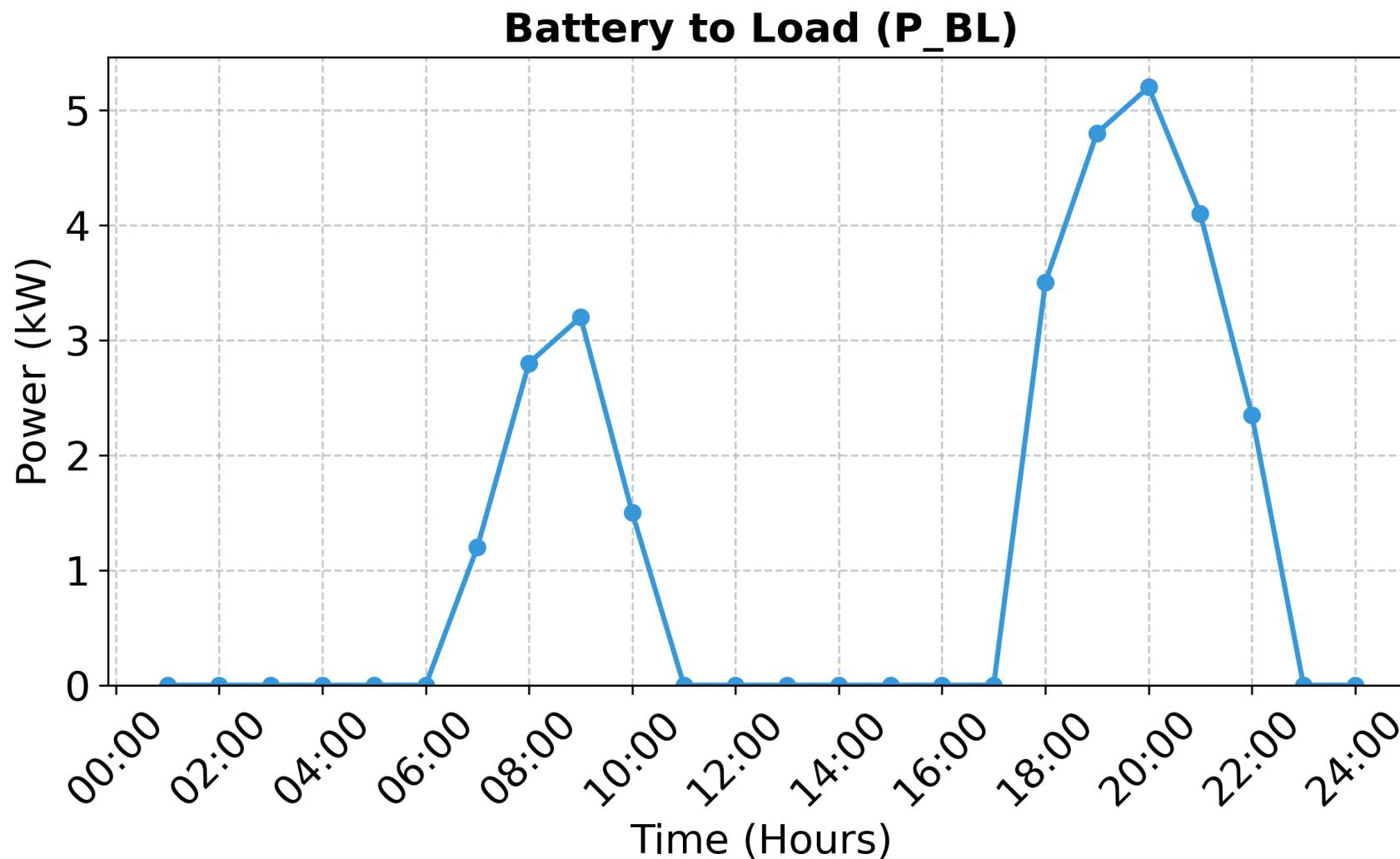


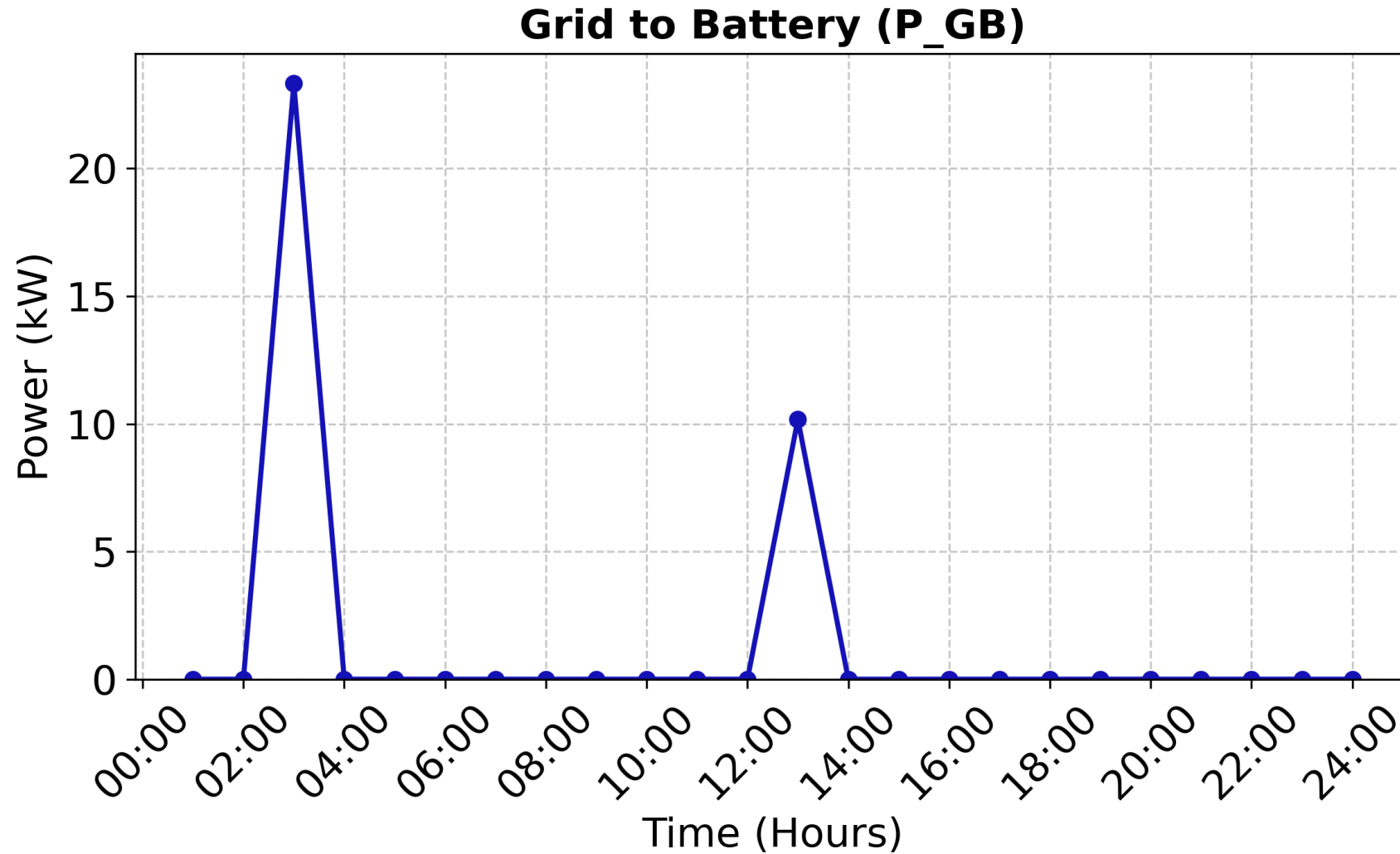








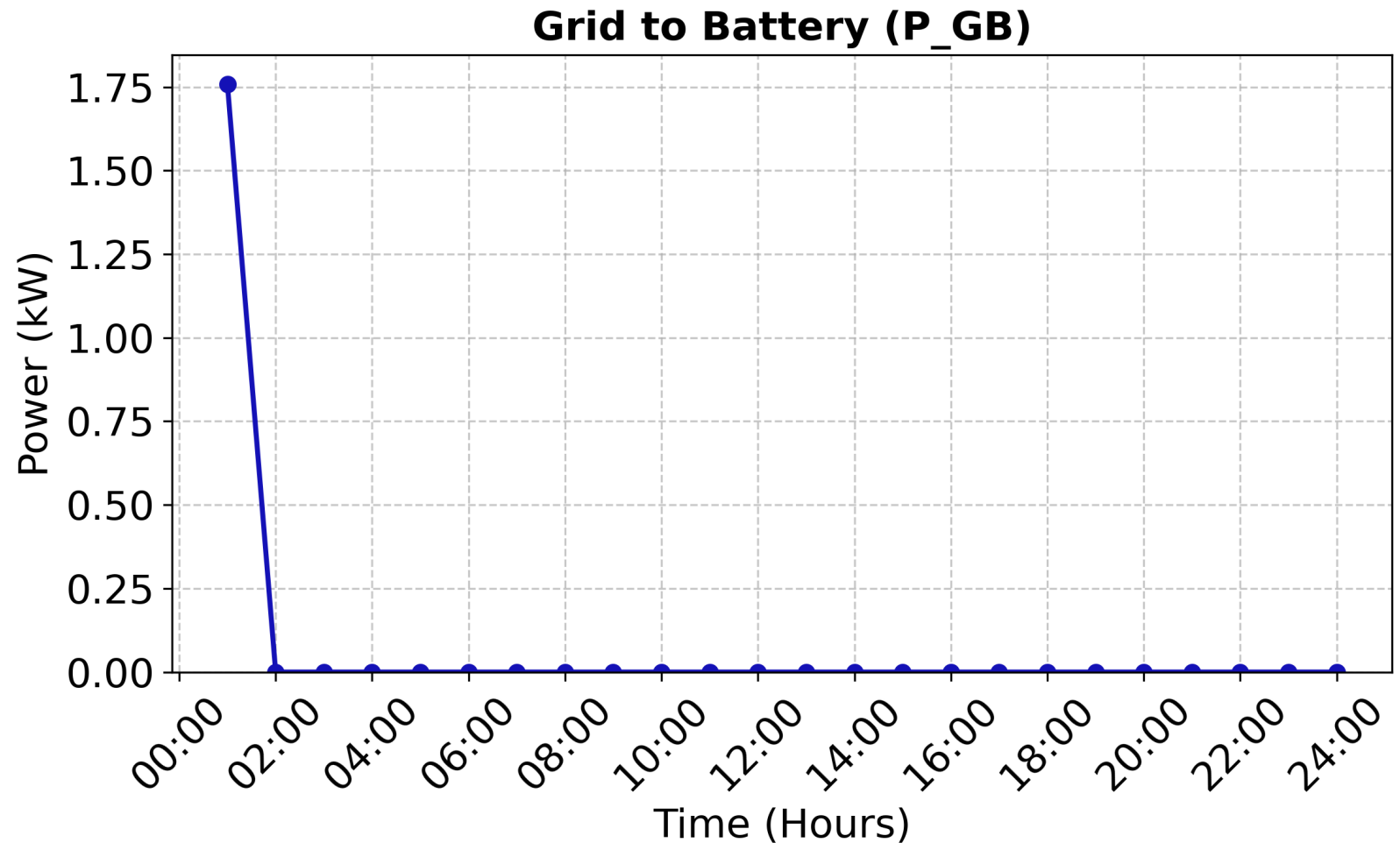


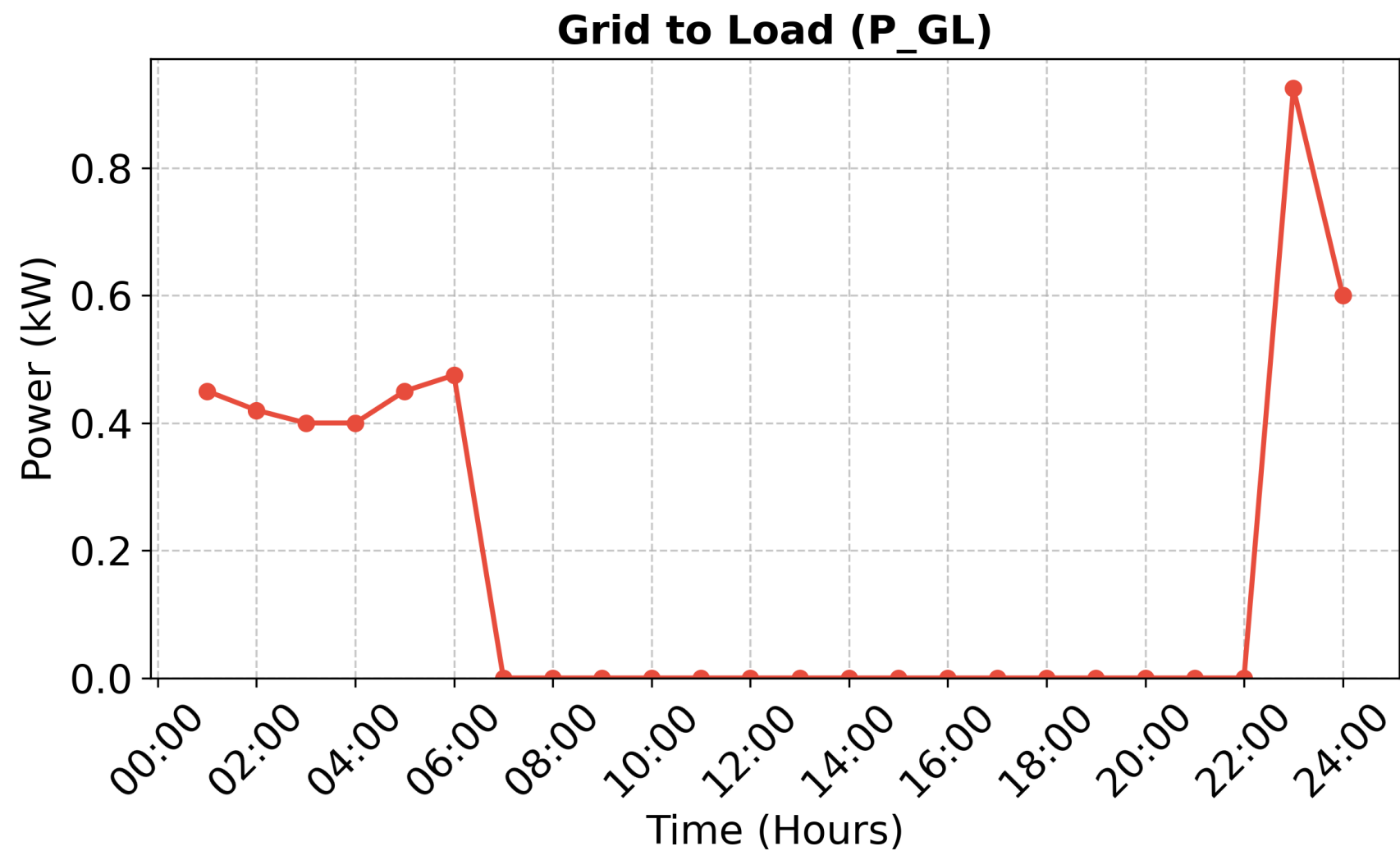


Smart Charging: Grid-to-Battery (P_{GB})

Phase 3: Analysis

The system stores cheap energy to survive the expensive Morning peak.

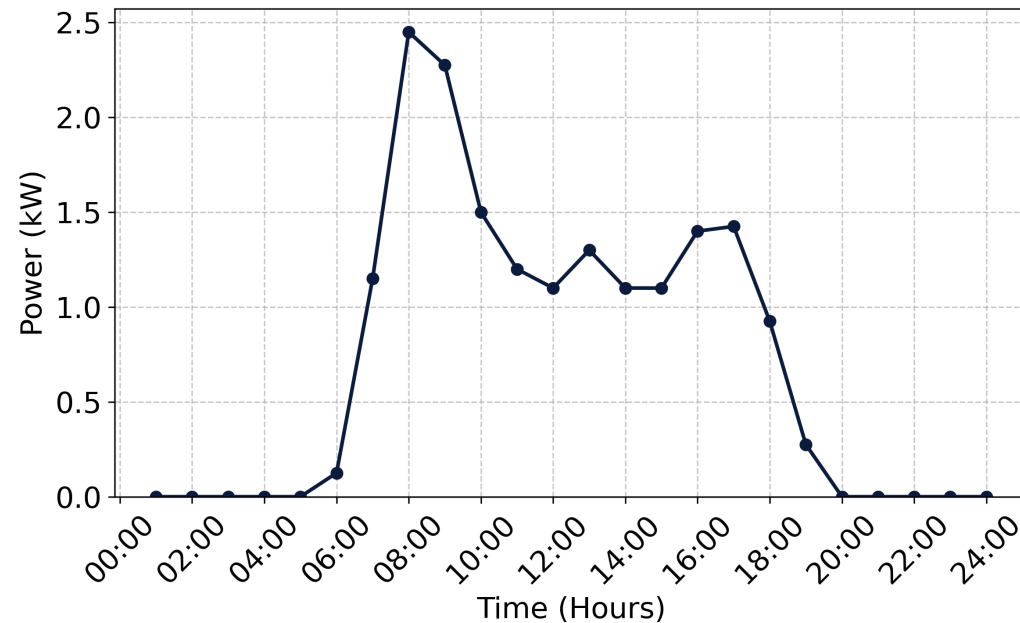




Utilizing Solar Generation

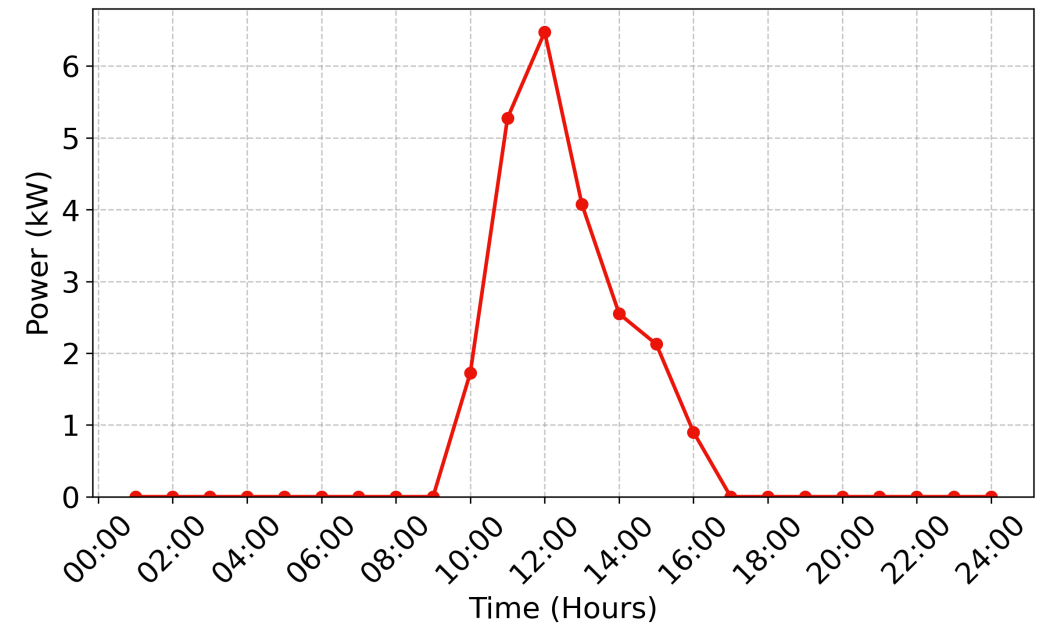
Phase 3: Analysis

Solar to Load (P_{SL})



During midday, solar covers the load.

Solar to Battery (P_{SB})

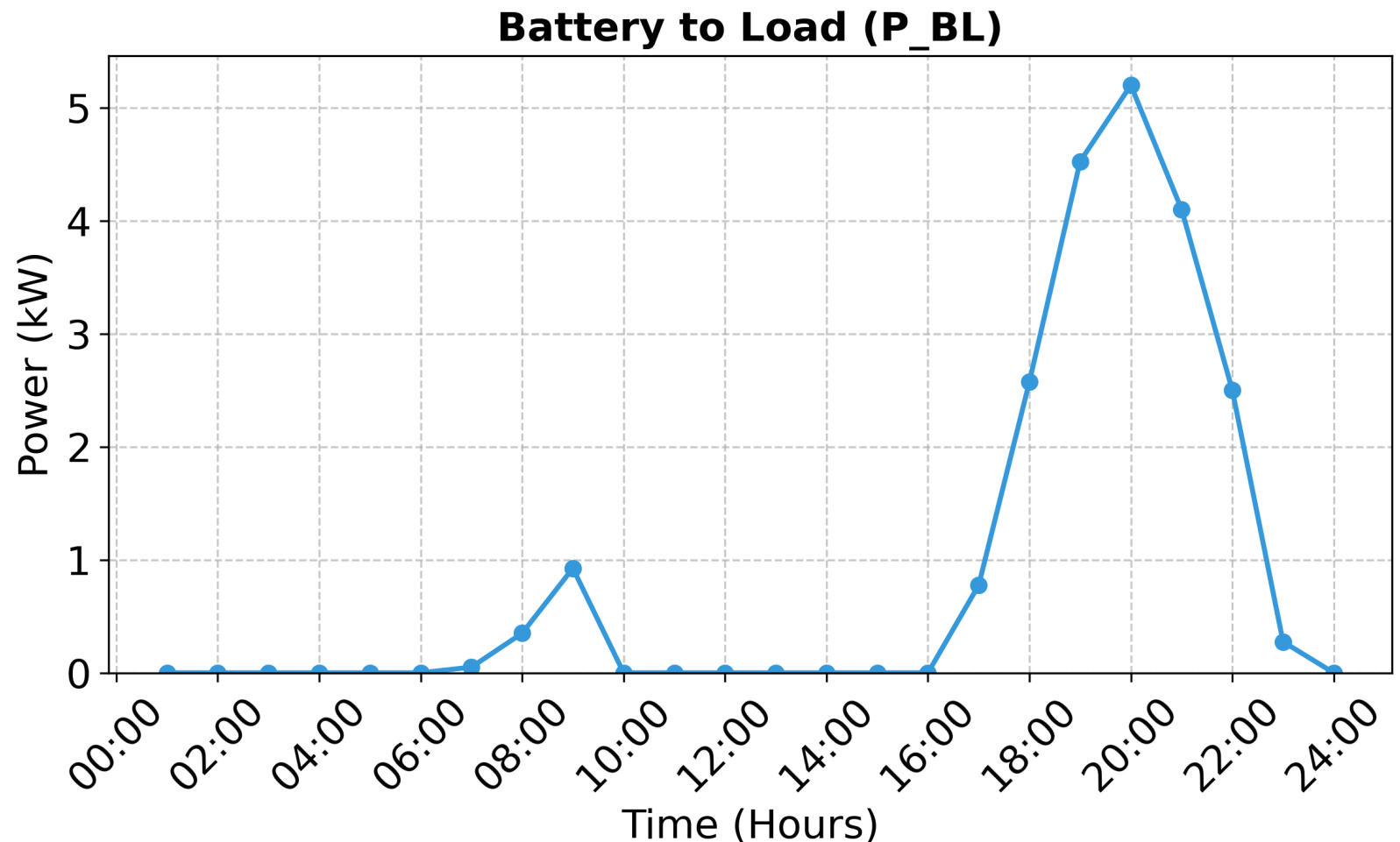


Excess solar fills the battery.

Avoiding Expensive Evening Rates (P_{BL})

Phase 3: Analysis

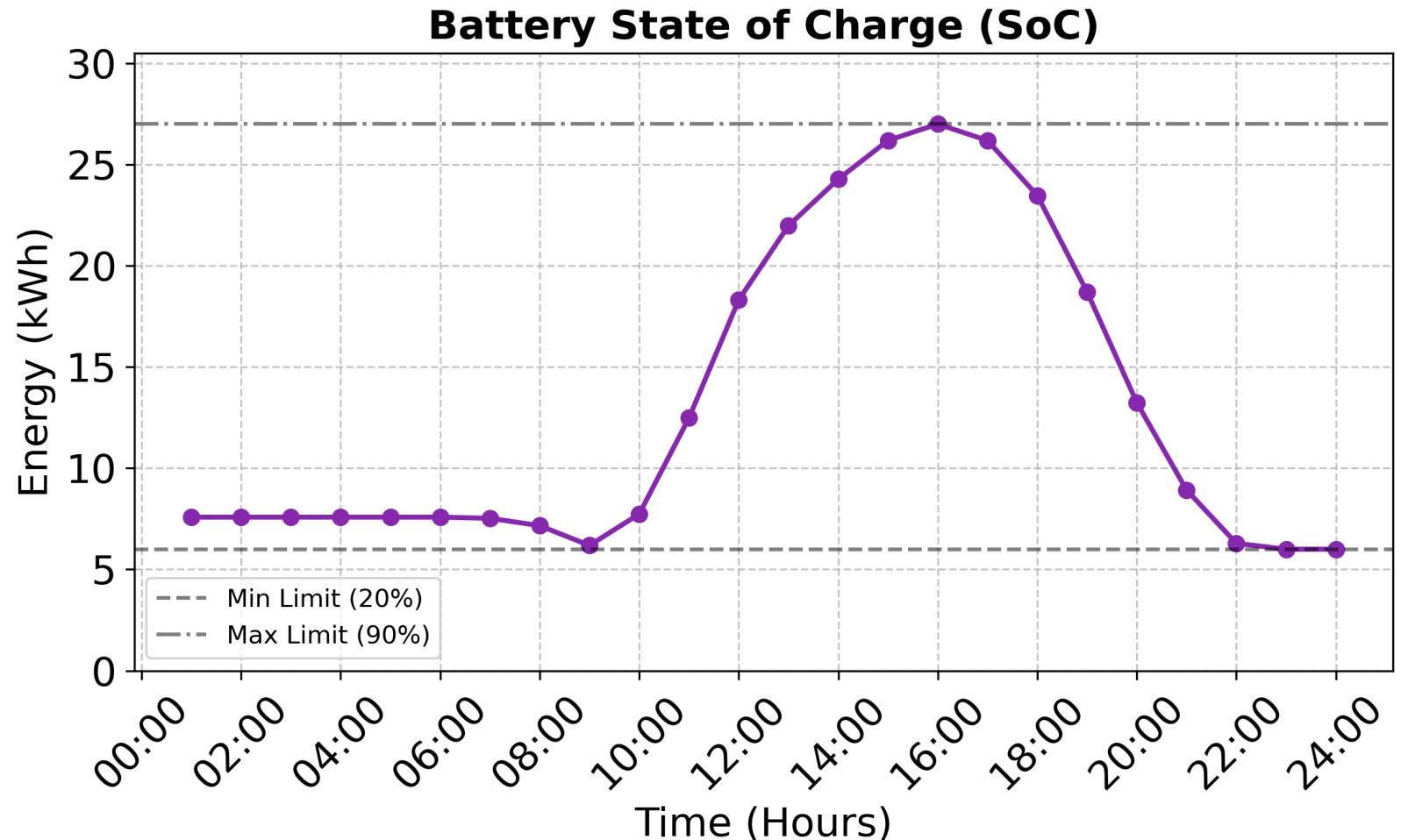
- During the evening peak (hours 18–21), dynamic prices hit their maximum (€0.48/kWh).
- Simultaneously, household load spikes.
- The optimal strategy is to aggressively discharge the battery (P_{BL}) to avoid buying expensive grid power.



Tracking Battery Energy (SoC)

Phase 3: Analysis

- The bounds strictly prevent the battery from dropping below the 20% safety threshold (6.0 kWh).
- **The Daily Cycle:**
 1. Morning pre-charge (Arbitrage).
 2. Solar top-off (Midday).
 3. Evening discharge (Peak Shaving).



Optimization Goal

Minimize electricity purchasing cost from the grid.

- Grid purchases shifted away from high-price periods.
- Solar energy utilization increased.
- Battery used to reduce expensive electricity imports.

Total Cost: without optimization

Total Cost = 11.74 euro

Cost on 30 kwh Battery backup: without solar power

Optimal Cost = 6.35 euro

Optimal Cost: with 30 kwh Battery, Solar power 25 kw

Optimal Cost = 0.81 euro

- Deterministic forecasts assumed.
- No uncertainty in PV generation.
- No uncertainty in load demand.
- No uncertainty in electricity prices.
- Battery and PV degradation not modeled.
- No export of excess energy to the grid.

Stochastic Optimization

- Multiple scenarios
- Probabilistic forecasts
- Expected cost minimization

Robust Optimization

- Worst-case analysis
- Uncertainty sets
- Guaranteed feasibility

Goal: Develop a controller that remains effective under forecast uncertainty.

Thank You
Questions?